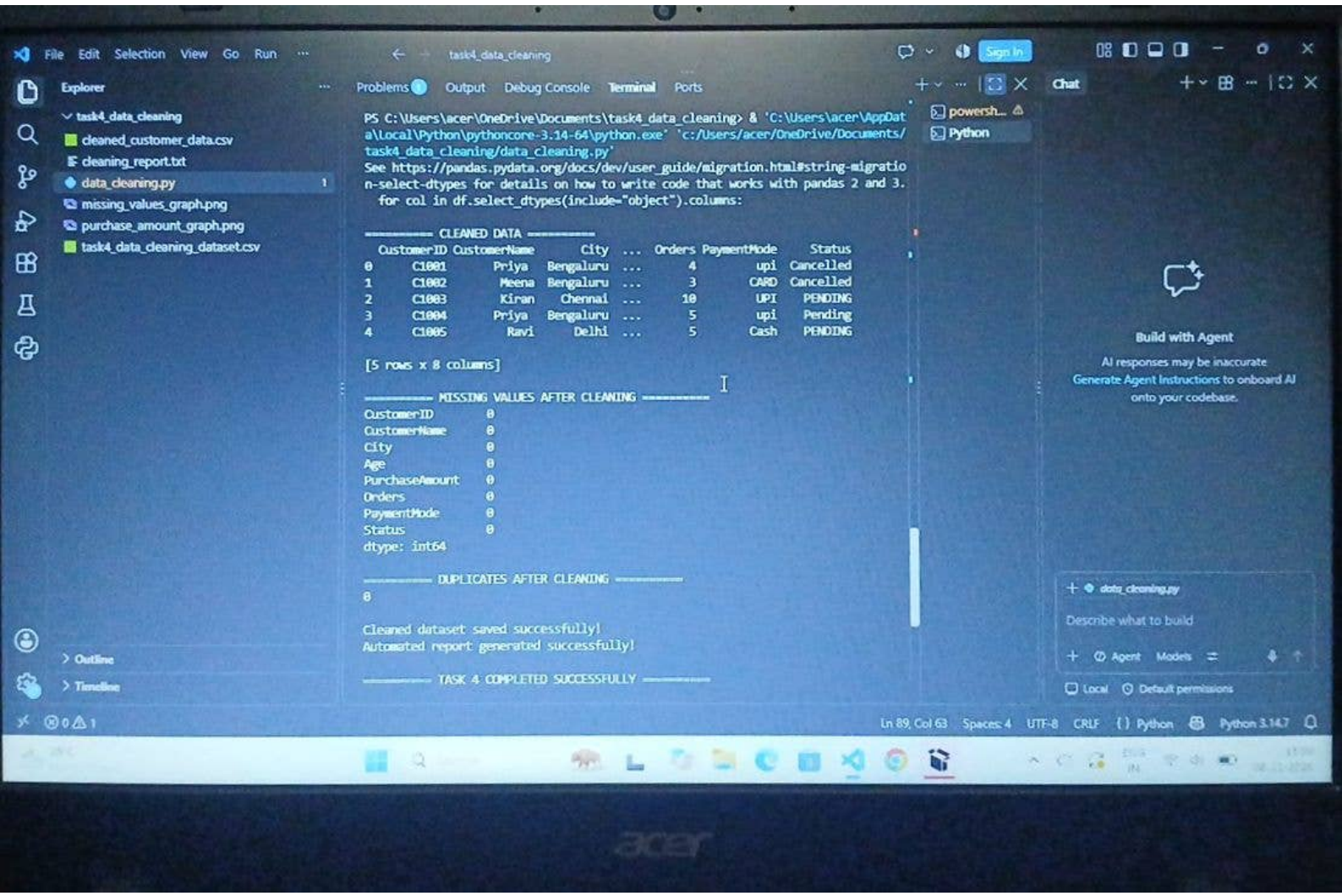
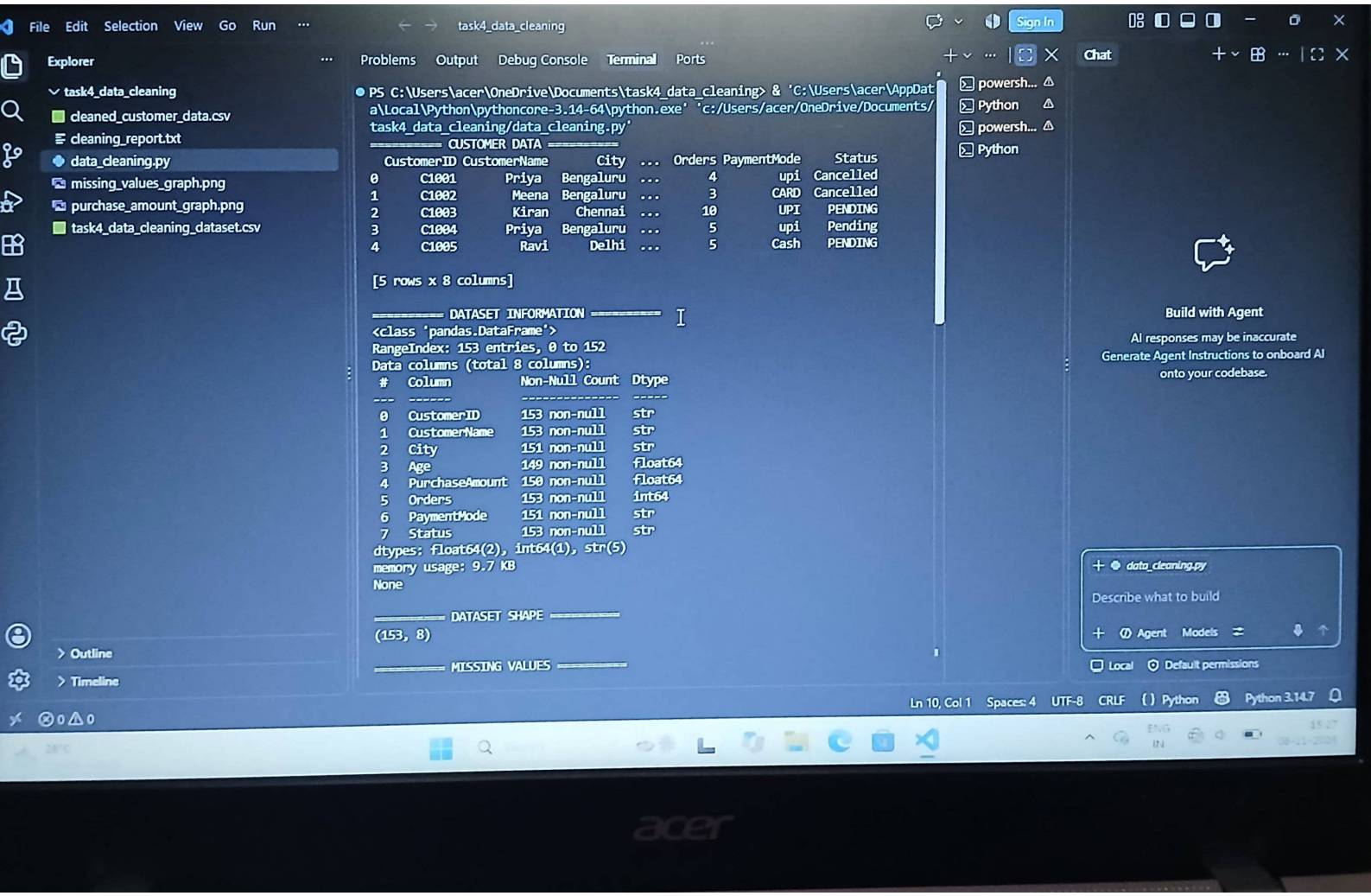
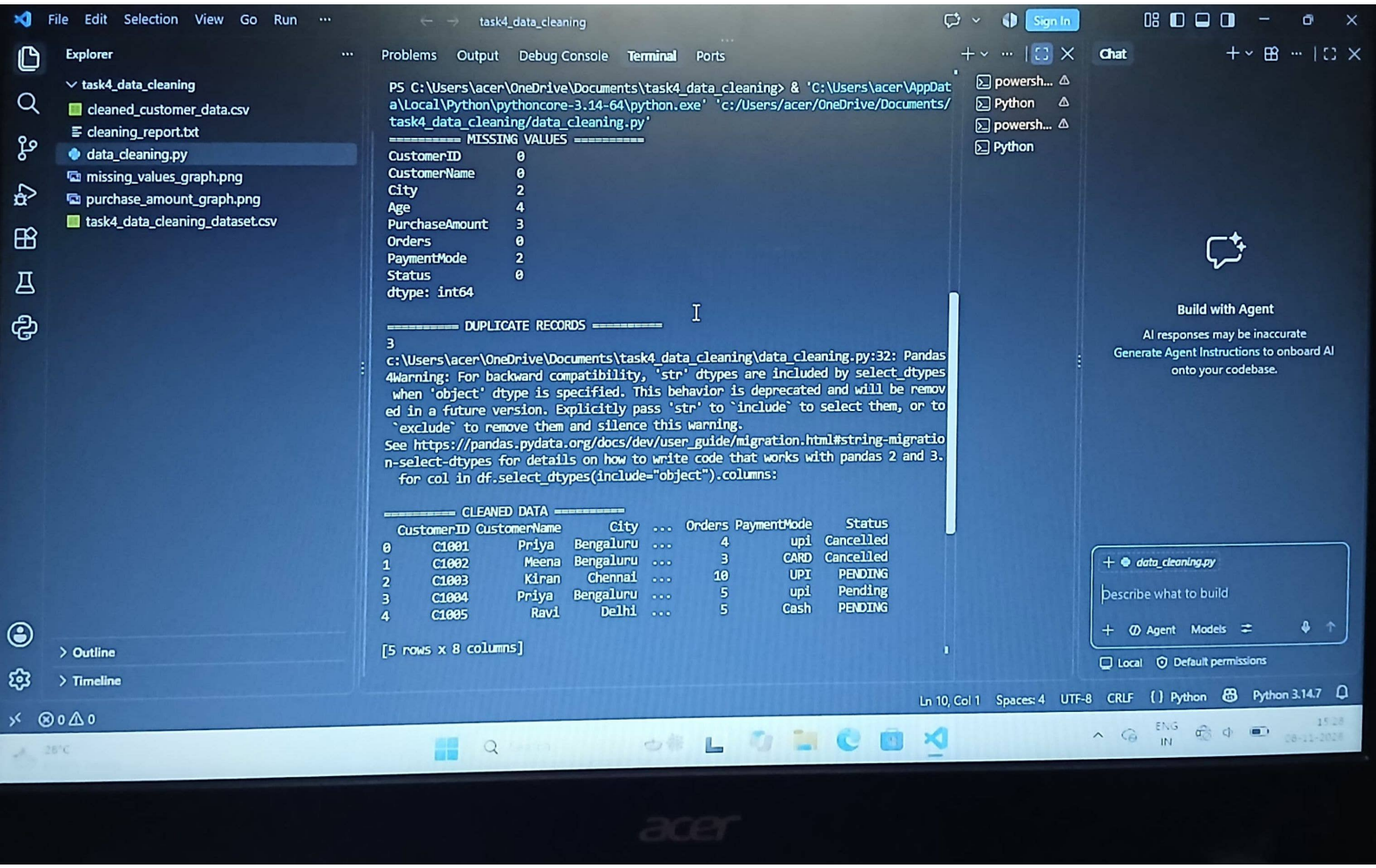
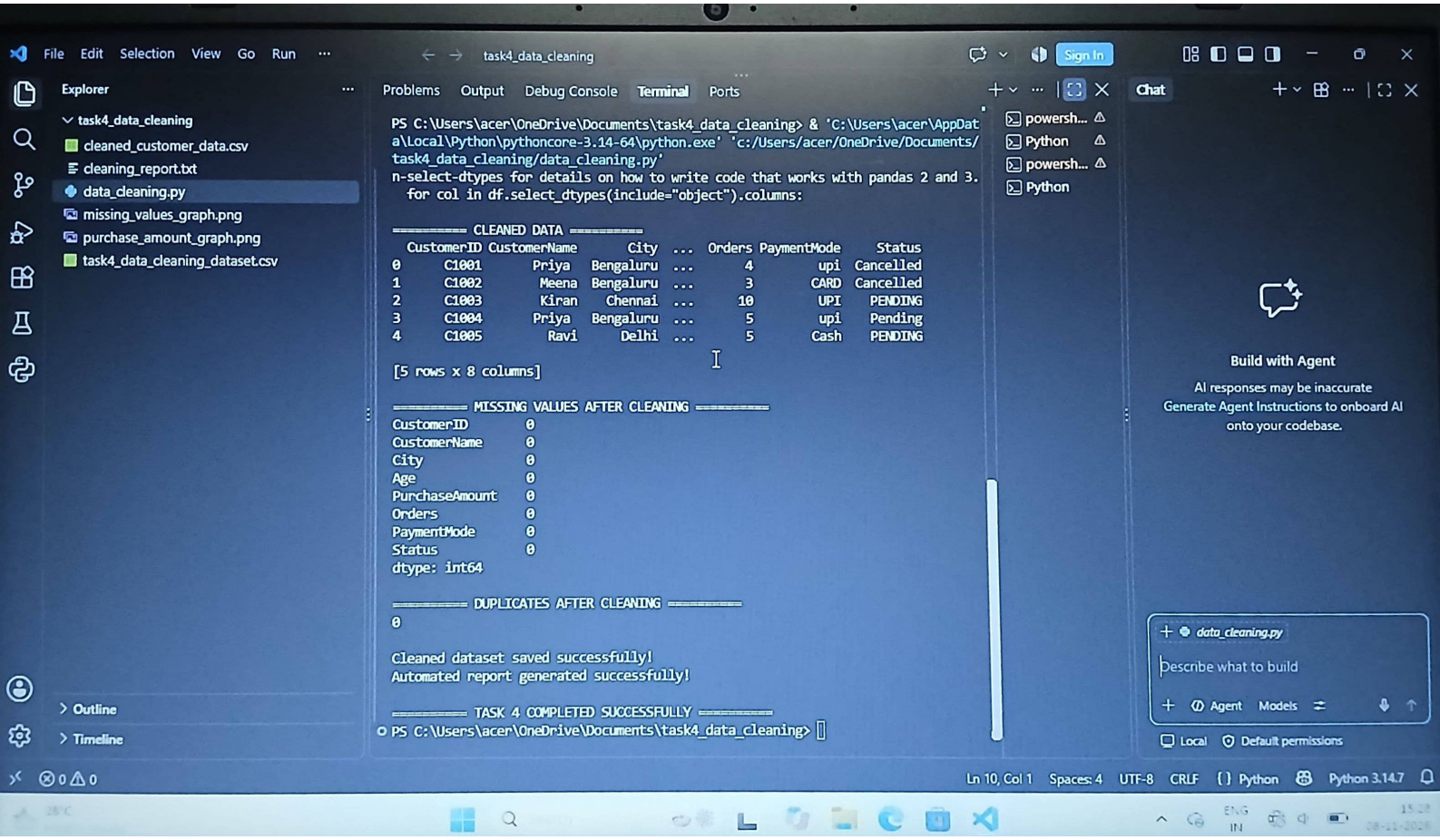

File Edit Selection View Go Run
task4_data_cleaning
Sign In
Explorer
task4_data_cleaning
cleaned_customer_data.csv
cleaning_report.txt
data_cleaning.py
missing_values_graph.png
purchase_amount_graph.png
task4_data_cleaning_dataset.csv
Problems Output Debug Console Terminal Ports
Chat
powersh...
Python
PS C:\Users\acer\OneDrive\Documents\task4_data_cleaning> & 'C:\Users\acer\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:/Users/acer/OneDrive/Documents/task4_data_cleaning/data_cleaning.py'
See https://pandas.pydata.org/docs/dev/user_guide/migration.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
for col in df.select_dtypes(include="object").columns:
========== CLEANED DATA ==========
CustomerID CustomerName City ... Orders PaymentMode Status
0 C1001 Priya Bengaluru ... 4 upi Cancelled
1 C1002 Meena Bengaluru ... 3 CARD Cancelled
2 C1003 Kiran Chennai ... 10 UPI PENDING
3 C1004 Priya Bengaluru ... 5 upi Pending
4 C1005 Ravi Delhi ... 5 Cash PENDING
[5 rows x 8 columns]
========== MISSING VALUES AFTER CLEANING ==========
CustomerID 0
CustomerName 0
City 0
Age 0
PurchaseAmount 0
Orders 0
PaymentMode 0
Status 0
dtype: int64
========== DUPLICATES AFTER CLEANING ==========
0
Cleaned dataset saved successfully!
Automated report generated successfully!
========== TASK 4 COMPLETED SUCCESSFULLY ==========
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
data_cleaning.py
Describe what to build
Agent Models
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Timeline
Ln 89, Col 63 Spaces: 4 UTF-8 CRLF Python Python 3.14.7


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PS C:\Users\acer\OneDrive\Documents\task4_data_cleaning> & 'C:\Users\acer\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:/Users/acer/OneDrive/Documents/task4_data_cleaning/data_cleaning.py'
========= CUSTOMER DATA =========
CustomerID CustomerName City ... Orders PaymentMode Status
0 C1001 Priya Bengaluru ... 4 upi Cancelled
1 C1002 Meena Bengaluru ... 3 CARD Cancelled
2 C1003 Kiran Chennai ... 10 UPI PENDING
3 C1004 Priya Bengaluru ... 5 upi Pending
4 C1005 Ravi Delhi ... 5 Cash PENDING
[5 rows x 8 columns]
========= DATASET INFORMATION =========
<class 'pandas.DataFrame'>
RangeIndex: 153 entries, 0 to 152
Data columns (total 8 columns):
Column Non-Null Count Dtype
0 CustomerID 153 non-null str
1 CustomerName 153 non-null str
2 City 151 non-null str
3 Age 149 non-null float64
4 PurchaseAmount 150 non-null float64
5 Orders 153 non-null int64
6 PaymentMode 151 non-null str
7 Status 153 non-null str
dtypes: float64(2), int64(1), str(5)
memory usage: 9.7 KB
None
========= DATASET SHAPE =========
(153, 8)
========= MISSING VALUES =========
powersh...
Python
powersh...
Python
Chat
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========= MISSING VALUES =========
CustomerID 0
CustomerName 0
City 2
Age 4
PurchaseAmount 3
Orders 0
PaymentMode 2
Status 0
dtype: int64
========= DUPLICATE RECORDS =========
3
c:\Users\acer\OneDrive\Documents\task4_data_cleaning\data_cleaning.py:32: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to `include` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/dev/user_guide/migration.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
for col in df.select_dtypes(include="object").columns:
========= CLEANED DATA =========
CustomerID CustomerName City ... Orders PaymentMode Status
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3 C1004 Priya Bengaluru ... 5 upi Pending
4 C1005 Ravi Delhi ... 5 Cash PENDING
[5 rows x 8 columns]
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0 0


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4 C1005 Ravi Delhi ... 5 Cash PENDING
[5 rows x 8 columns]
========= MISSING VALUES AFTER CLEANING =========
CustomerID 0
CustomerName 0
City 0
Age 0
PurchaseAmount 0
Orders 0
PaymentMode 0
Status 0
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========= DUPLICATES AFTER CLEANING =========
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Cleaned dataset saved successfully!
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========= TASK 4 COMPLETED SUCCESSFULLY =========
PS C:\Users\acer\OneDrive\Documents\task4_data_cleaning>
powersh...
Python
powersh...
Python
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Ln 10, Col 1 Spaces: 4 UTF-8 CRLF Python Python 3.14.7

File Edit Selection View Go Run ··· task4_data_cleaning Sign In

Explorer

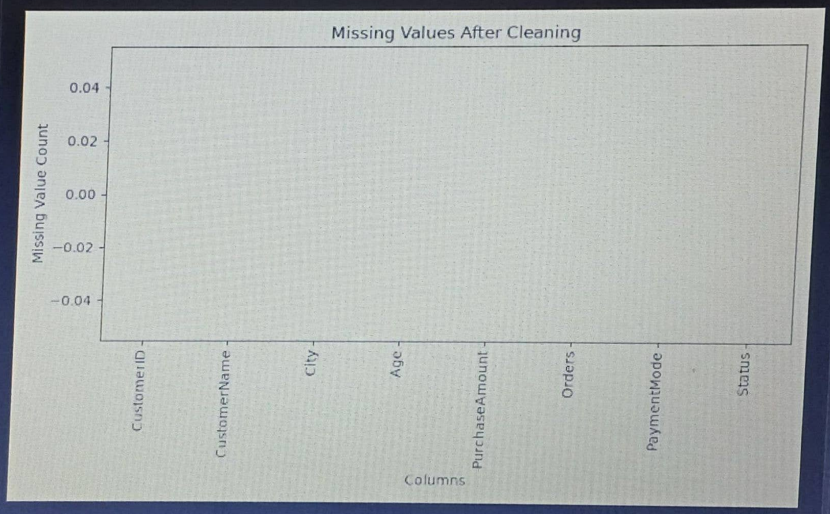
task4_data_cleaning

- cleaned_customer_data.csv
- cleaning_report.txt
- data_cleaning.py
- missing_values_graph.png
- purchase_amount_graph.png
- task4_data_cleaning_dataset.csv

data_cleaning.py | missing_values_graph.png

missing_values_graph.png

Image Preview



Chat

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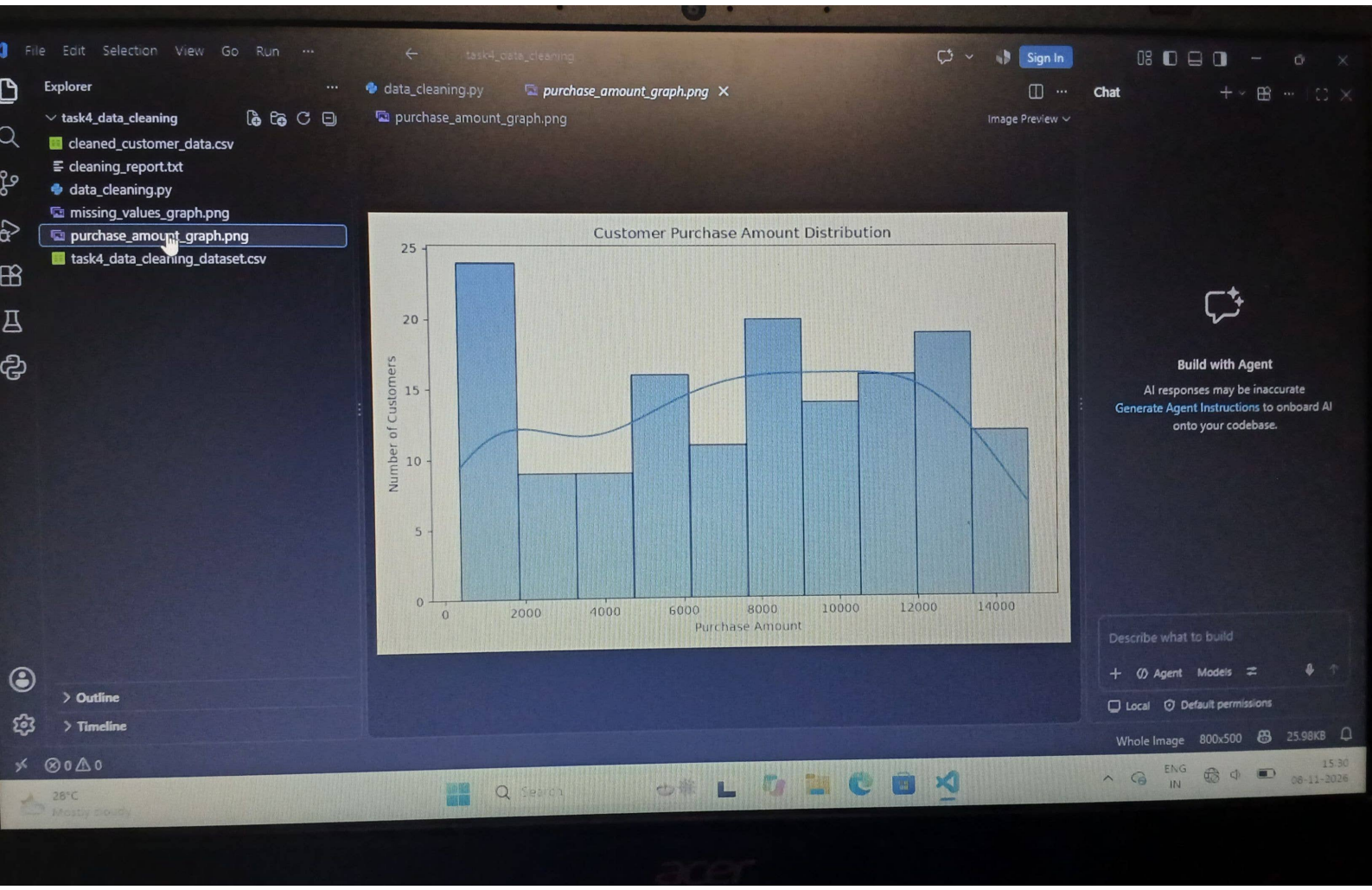
0 0

84% 800x500 22.85KB

28°C Mostly cloudy

Search

ENG IN 15:30 08-11-2026


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task4_data_cleaning
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Image Preview
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data_cleaning.py
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purchase_amount_graph.png
task4_data_cleaning_dataset.csv
Customer Purchase Amount Distribution
Number of Customers
Purchase Amount
0
5
10
15
20
25
0
2000
4000
6000
8000
10000
12000
14000
Build with Agent
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Describe what to build
Agent
Models
Local
Default permissions
Outline
Timeline
Whole Image
800x500
25.98KB
28°C
Mostly cloudy
Search
ENG
IN
15:30
08-11-2026